

Ordering, adding and subtracting decimals

A Cambridge insert: place value decides the order, and the points line up for the arithmetic.

LESSON 76–77

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 4.1–4.2

LESSON CLOCK

20'

25'

25'

8'

2

Ordering	20 min
Adding	25 min
Subtracting	25 min
Estimating	8 min
Homework	2 min

LEARNING OBJECTIVES

- Compare and order decimals of different lengths.
- Add and subtract decimals by lining up the points.
- Use zeros to fill empty places.
- Estimate a decimal calculation before doing it.

TEXTBOOK

National	pp. 211–216
Cambridge	Stage 7, pp. 42–48
Workbook	Exercise 4.1

01

Explanation

Ordering

Compare column by column from the left, not by counting digits. Fill the short numbers with trailing zeros so every number has the same length.

NUMBERS	FILLED OUT	ORDER
0.7, 0.65	0.70, 0.65	0.65 < 0.7
3.4, 3.25, 3.5	3.40, 3.25, 3.50	3.25 < 3.4 < 3.5
0.09, 0.1	0.09, 0.10	0.09 < 0.1

MORE DIGITS DOES NOT MEAN BIGGER

0.65 has more digits than 0.7 and is smaller. The comparison is made column by column: tenths first, then hundredths.

Adding

Write the numbers with their points in a vertical line and add column by column, exactly as with whole numbers.

SUM	LINED UP	ANSWER
$4.7 + 2.85$	$4.70 + 2.85$	7.55
$12.6 + 0.47$	$12.60 + 0.47$	13.07
$3 + 0.45$	$3.00 + 0.45$	3.45
$0.8 + 0.9$	$0.8 + 0.9$	1.7

A WHOLE NUMBER HAS A POINT TOO

3 is 3.00 when it is being added to hundredths. Writing the zeros makes the columns line up and the arithmetic ordinary.

Subtracting

DIFFERENCE	LINED UP	ANSWER
$9.1 - 3.62$	$9.10 - 3.62$	5.48
$5 - 1.35$	$5.00 - 1.35$	3.65
$7.4 - 0.85$	$7.40 - 0.85$	6.55
$0.5 - 0.28$	$0.50 - 0.28$	0.22

FILL THE GAP BEFORE BORROWING

$9.1 - 3.62$ needs the 9.1 written as 9.10; without the zero there is nothing in the hundredths column to borrow from, and the answer comes out wrong by a tenth.

Estimating

CALCULATION	ESTIMATE	EXACT
$4.7 + 2.85$	$5 + 3 = 8$	7.55
$9.1 - 3.62$	$9 - 4 = 5$	5.48
$12.6 + 0.47$	$13 + 0 = 13$	13.07

THE ESTIMATE PLACES THE POINT

An answer of 75.5 or 0.755 would be rejected at once by an estimate of 8 . A misplaced decimal point is the error estimating is best at catching.

02 Worked examples

EXAMPLE 1 Order 3.4, 3.25 and 3.5.

- 1

Fill out: 3.40, 3.25, 3.50.
- 2

Compare the tenths: $2 < 4 < 5$.
- 3

$3.25 < 3.4 < 3.5$

ANSWER

$3.25 < 3.4 < 3.5$

EXAMPLE 2 Compute $9.1 - 3.62$.

- ① Write 9.1 as 9.10 .
- ② Line up the points and subtract.
- ③ $= 5.48$
Estimate: $9 - 4 = 5$ ✓

ANSWER

 5.48 **EXAMPLE 3** Compute $3 + 0.45$ and $5 - 1.35$.

- ① $3.00 + 0.45 = 3.45$
- ② $5.00 - 1.35 = 3.65$
Whole numbers get zeros too.

ANSWER

 3.45 and 3.65 **03** Interactive model

Ask which is bigger, 0.7 or 0.65 , before teaching anything; the split in the class shows exactly who is counting digits rather than reading place value.

Place value decides

Which is larger, 0.7 or 0.65 ?

Which is larger, 0.09 or 0.1 ?

$4.7 + 2.85$ equals:

$9.1 - 3.62$ equals:

5.48

6.48

5.52

6.52

$3 + 0.45$ equals:

0.75

3.45

7.5

3.045

$5 - 1.35$ equals:

3.65

4.65

3.75

4.35

$0.8 + 0.9$ equals:

0.17

1.7

1.07

0.98

An estimate of $4.7 + 2.85$ is:

8

80

0.8

7.55

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Decimal	O‘nli kasr	Десятичная дробь
Decimal point	O‘nli nuqta (vergul)	Десятичная запятая
Tenth	O‘ndan bir	Десятая
Hundredth	Yuzdan bir	Сотая
Thousandth	Mingdan bir	Тысячная
To order	Tartiblash	Упорядочить
Trailing zero	Oxirgi nol	Конечный ноль
To line up	Ustma-ust qo‘yish	Выровнять

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To compare decimals you:

count the digits

compare column by column

add them

round them

0.65 compared with 0.7 is:

larger

smaller

equal

undetermined

To add decimals you line up:

the last digits

the decimal points

the first digits

nothing

9.1 written to two decimal places is:

9.01

9.10

9.1

0.91

$12.6 + 0.47$ equals:

13.07

12.13

17.6

13.7

An estimate mainly checks:

the digits

the decimal point

the signs

the units

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Larger: 0.7 or 0.65

ANSWER 0.7

2. Larger: 0.09 or 0.1

ANSWER 0.1

3. Order 3.4 , 3.25 , 3.5

ANSWER $3.25 < 3.4 < 3.5$

4. $4.7 + 2.85$

ANSWER 7.55

5. $12.6 + 0.47$

ANSWER 13.07

6. $3 + 0.45$

ANSWER 3.45

7. $0.8 + 0.9$

ANSWER 1.7

Medium · the standard question

7 PROBLEMS

1. $9.1 - 3.62$

ANSWER 5.48

2. $5 - 1.35$

ANSWER 3.65

3. $7.4 - 0.85$

ANSWER 6.55

4. $0.5 - 0.28$

ANSWER 0.22

5. Estimate $4.7 + 2.85$

ANSWER 8

6. Estimate $9.1 - 3.62$

ANSWER 5

7. Order $0.5, 0.45, 0.505$

ANSWER $0.45 < 0.5 < 0.505$

Hard · stretch and reason

7 PROBLEMS

1. $12.05 + 3.7 + 0.68$

ANSWER 16.43

2. $20 - 6.47$

ANSWER 13.53

3. $4.2 - 1.35 + 0.7$

ANSWER 3.55

4. Order 1.1 , 1.01 , 1.011 , 1.101

ANSWER $1.01 < 1.011 < 1.1 < 1.101$

5. The number halfway between 0.6 and 0.7

ANSWER 0.65

6. A length of 2.4 m less 85 cm

ANSWER 1.55 m

7. Why write 9.1 as 9.10 before subtracting?

ANSWER The hundredths column needs a digit to borrow from

07 Homework

Homework — 5 tasks

Fill every number with trailing zeros so all have the same length.

1. Order 2.3 , 2.25 , 2.4 and 2.05 .
2. Compute $6.8 + 3.45$ and $0.9 + 0.75$.
3. Compute $8.2 - 4.67$ and $6 - 2.48$.
4. Estimate each of the four answers above and compare with your exact values.
5. A plank 3.5 m long has 68 cm cut off. How much is left, in metres?